

MALAYBALAY, Philippines

WMO: 987510

Lat: 8.1514N

Lon: 125.1339E

Elev: 2057

StdP: 13.64

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	63.6	65.2	61.1	86.7	67.4	62.8	92.2	68.2	9.8	80.5	7.8	80.6	0.4	30	0.178

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	16.7	89.9	78.1	88.7	77.8	87.6	77.5	80.8	87.3	80.0	86.5	79.1	85.6	3.0	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.0	162.6	86.3	78.0	157.3	85.4	77.1	152.7	84.6	46.5	87.5	45.4	86.7	44.4	85.8	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.5	5.7	5.1	60.2	92.6	1.9	1.7	58.8	93.7	57.7	94.7	56.6	95.6	55.2	96.8		
			59.0	83.3	1.9	0.9	57.7	83.9	56.6	84.5	55.5	85.0	54.2	85.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	76.5	75.2	75.6	76.5	77.7	78.3	77.0	75.9	76.1	76.1	76.6	76.3
		DBStd	2.14	2.24	2.09	1.96	2.06	1.84	1.98	1.94	1.75	1.78	1.92	2.11
		HDD50	0	0	0	0	0	0	0	0	0	0	0	0
		HDD65	0	0	0	0	0	0	0	0	0	0	0	0
		CDD50	9675	781	716	823	832	876	811	804	810	784	825	797
		CDD65	4200	316	296	358	382	411	361	339	345	334	360	347
		CDH74	25699	1744	1895	2597	2976	2882	2134	1710	1798	1711	1999	2110
		CDH80	7406	433	539	897	1119	980	575	359	401	390	526	601
(14)	Wind	WSAvg	1.3	1.3	1.4	1.4	1.3	1.2	1.2	1.4	1.4	1.2	1.1	1.3
(15)	Precipitation	PrecAvg	103.6	7.2	5.4	5.0	5.2	9.7	12.0	12.6	11.3	11.8	9.8	6.1
(16)		PrecMax	141.7	16.1	11.3	13.3	22.3	19.0	21.3	20.6	18.5	18.0	20.4	13.5
(17)		PrecMin	75.5	2.0	1.0	0.9	0.9	3.5	8.3	6.9	5.3	5.0	2.0	2.6
(18)		PrecStd	17.4	4.1	2.9	2.8	4.6	3.4	2.9	3.6	3.6	3.3	3.9	3.3
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	87.9	89.0	90.7	91.8	91.0	89.1	87.8	87.9	88.5	89.4	89.1	88.9
(20)		0.4% MCWB	76.9	77.0	76.2	78.3	78.8	78.7	78.5	78.0	78.2	78.7	78.5	78.4
(21)		2% DB	85.9	86.9	88.7	89.8	89.1	87.0	85.6	85.9	86.1	87.0	87.2	86.9
(22)		2% MCWB	76.2	76.4	76.4	77.9	78.7	78.2	77.6	77.3	77.6	78.0	77.7	77.4
(23)		5% DB	84.4	85.4	87.0	88.2	87.5	85.4	83.8	84.3	84.3	85.2	85.7	85.4
(24)		5% MCWB	75.6	75.7	76.1	77.4	78.3	77.6	76.8	76.7	77.0	77.4	77.3	76.7
(25)		10% DB	82.6	83.8	85.4	86.4	85.7	83.6	82.0	82.4	82.4	83.3	83.9	83.8
(26)		10% MCWB	75.0	75.0	75.7	76.8	77.6	76.9	76.0	76.0	76.2	76.6	76.5	76.0
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	79.9	80.6	80.9	81.7	81.8	80.7	80.0	80.0	80.3	80.8	80.8	80.6
(28)		0.4% MCDB	85.6	87.1	87.8	89.0	88.6	86.7	85.9	85.8	86.3	86.9	87.0	86.7
(29)		2% WB	77.9	78.5	79.1	80.1	80.1	79.2	78.3	78.3	78.6	79.1	79.2	78.8
(30)		2% MCDB	84.0	84.9	86.2	87.6	87.1	85.4	84.1	84.2	84.6	85.4	85.6	85.4
(31)		5% WB	76.6	77.0	77.6	78.7	78.9	78.1	77.2	77.2	77.4	77.8	78.0	77.4
(32)		5% MCDB	82.9	83.6	85.0	86.3	86.0	84.2	82.8	83.0	83.2	84.0	84.3	83.8
(33)		10% WB	75.4	75.6	76.3	77.2	77.7	77.0	76.1	76.1	76.3	76.7	76.7	76.4
(34)		10% MCDB	81.3	82.1	83.6	85.0	84.6	83.0	81.3	81.5	81.7	82.6	82.9	82.5
(35)	Mean Daily Temperature Range	MDBR	14.9	16.9	18.7	18.9	16.7	14.4	13.0	13.8	13.9	14.7	15.8	15.7
(36)		5% DB MCDBR	19.0	20.3	22.2	21.9	19.4	17.1	16.3	17.0	17.1	18.0	19.0	18.6
(37)		5% DB MCWBR	10.7	11.6	12.2	11.7	10.5	9.6	9.4	9.6	9.9	10.3	10.8	10.4
(38)		5% WB MCDBR	17.1	18.5	20.0	20.2	18.0	16.3	15.2	15.5	15.9	16.8	17.9	17.2
(39)		5% WB MCWBR	10.9	12.1	12.9	12.8	10.9	9.9	9.5	9.7	10.0	10.6	11.6	10.8
(40)	Clear-Sky Solar Irradiance	taub	0.378	0.376	0.377	0.386	0.394	0.385	0.388	0.408	0.407	0.417	0.383	0.379
(41)		taud	2.521	2.504	2.512	2.479	2.482	2.513	2.499	2.419	2.413	2.385	2.506	2.518
(42)		Ebn at Noon	297	300	299	292	284	283	283	282	286	283	293	294
(43)		Edh at Noon	34	35	35	36	35	33	34	38	38	39	34	33
(44)	All-Sky Solar Radiation	RadAvg	1270	1477	1661	1760	1673	1571	1547	1604	1597	1537	1405	1321
(45)		RadStd	150	151	134	111	108	80	90	103	110	104	125	105

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.43	N/A	N/A	N/A

Nomenclature: See separate page